
Week 36: Long Run Behavior of Markov Chains (Part 1)

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These notes are written by myself, and errors may and will occur. When in doubt, trust the book and Gunnars lectures!

1. Theory

We are now interested in exploring the long-term behavior of Markov chains, meaning determining the limit

$$\lim_{t \rightarrow \infty} P_{i,j}^{(t)} = \lim_{t \rightarrow \infty} P(X_t = j \mid X_0 = i).$$

For this we have to introduce some additional terminology and definitions.

Definition 1.1. (*Limiting distribution*) Consider a DT-MC $\{X_t\}$. We call $\pi = (\pi_0, \pi_1, \dots)^\top$ the *limiting distribution* if the following two conditions are satisfied:

1. The limits $\pi_j = \lim_{t \rightarrow \infty} P_{i,j}^{(t)}$ exist for $j = 0, 1, \dots$ and do not depend on i .
2. $\sum_{j=0}^{\infty} \pi_j = 1$.

Notes:

- 1. $\implies$ 2. for finite state spaces $\{0, 1, \dots, N\}$, but not necessarily for infinite state spaces.
- π_j can be interpreted as the probability of being in state j after many transitions.
- I let π be a column vector, but some authors let it be a row vector. The two conventions are equivalent, but resulting matrix equations look different.

Definition 1.2. (*Regular Markov chain*) Consider a discrete-time Markov chain (DT-MC) $\{X_t\}$ with finite state space $\{0, 1, \dots, N\}$ and transition probability matrix $\mathbf{P}$. If there exists a positive integer $k > 0$ so that all elements of $\mathbf{P}^k$ are strictly positive, we call $\mathbf{P}$ and $\{X_t\}$ *regular*.

Written more mathematically, we have $\mathbf{P}$ regular $\iff \exists k > 0$ s.t. $P_{i,j}^{(k)} > 0$ for all $i, j \in S$.

Theorem 1.3. Let $\{X_t\}$ be a regular DT-MC with finite state space $\{0, 1, \dots, N\}$ and transition probability matrix $\mathbf{P}$. Then there exists a unique limiting distribution π , which is the non-negative solution of

$$\begin{cases} \pi = \mathbf{P}^\top \pi \\ \mathbf{1}^\top \pi = 1 \end{cases}$$

Note:

- Regularity is a sufficient condition for the existence of a unique limiting distribution.

Definition 1.4. (*Stationary distribution*) A probability distribution $\pi = (\pi_0, \pi_1, \dots)^\top$ is called a *stationary distribution* if

$$\pi_j = \sum_{i=0}^{\infty} \pi_i P_{i,j} \quad \text{for } j = 0, 1, \dots \quad \text{and} \quad \sum_{j=0}^{\infty} \pi_j = 1$$

or, equivalently, $\pi = \mathbf{P}^\top \pi$ and $\mathbf{1}^\top \pi = 1$ for finite state spaces.

Notes:

- If $X_0 \sim \pi$, then $X_t \sim \pi$ for every $t \geq 0$.
- Limiting distribution $\implies$ Stationary distribution (but not the other way around).

Definition 1.5. (*Doubly stochastic*) The transition probability matrix $\mathbf{P}$ is called *doubly stochastic* if

$$\sum_{k=0}^N P_{i,k} = \sum_{k=0}^N P_{k,j} = 1$$

for all states i and j .

Example 1.6. The $\mathbf{C}$ from [Problem 1.1](#) is doubly stochastic. Not every transition probability matrix is doubly stochastic, its rows always sum to one, but its columns need not.

Theorem 1.7. *Let the Markov chain $\{X_t : t = 0, 1, \dots\}$ be regular with finite state space $\{0, 1, \dots, N\}$. If the transition probability matrix $\mathbf{P}$ is doubly stochastic, then the limiting distribution is uniform*

$$\pi = \left(\frac{1}{N+1}, \frac{1}{N+1}, \dots, \frac{1}{N+1} \right)^\top.$$

Example 1.8. (*Doubly stochastic matrices*) Consider the transition probability matrix used in [Problem 1.2](#)

$$\mathbf{P} = \begin{bmatrix} p & q & 0 \\ 0 & p & q \\ q & 0 & p \end{bmatrix},$$

and assume that $0 < p, q < 1$ and $p + q = 1$ to avoid the edge-cases. We already showed that this Markov chain is regular. It is doubly stochastic because the rows and columns sum to one. Hence, the limiting distribution is uniform $\pi = (1/3, 1/3, 1/3)^\top$.

Theorem 1.9. (*Long-run fraction of time*) *In a regular Markov chain $\{X_t : t = 0, 1, \dots\}$, the limiting distribution $\pi = (\pi_0, \pi_1, \dots, \pi_N)^\top$ gives the long-run fraction of time spent in each state. I.e.,*

$$\pi_j = \lim_{n \rightarrow \infty} \mathbb{E} \left[\frac{1}{n} \sum_{k=0}^{n-1} \mathbb{1}\{X_k = j\} \mid X_0 = i \right]$$

for any state i

Problem 1.1. Are

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \text{and/or} \quad \mathbf{C} = \begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 1/2 & 1/2 \end{bmatrix}$$

regular?

Solution. For $\mathbf{A}$, we have $\mathbf{A}^k = \mathbf{A}$ for all k , and therefore $\mathbf{A}$ is not regular. For $\mathbf{B}$, we have

$$\mathbf{B}^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad \mathbf{B}^3 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix},$$

so $\mathbf{B}$ is not regular either. For $\mathbf{C}$, we have

$$\mathbf{C}^2 = \begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \end{bmatrix},$$

meaning that $\mathbf{C}$ is regular.

Problem 1.2. (Fixing the LF from Exercise 3) Consider the transition probability matrix

$$\mathbf{P} = \begin{bmatrix} p & q & 0 \\ 0 & p & q \\ q & 0 & p \end{bmatrix},$$

where $p, q \geq 0$ and $p + q = 1$. Consider the cases

- $0 < p < 1$ and $q = 1 - p > 0$
- $p = 1$ and $q = 0$.
- $p = 0$ and $q = 1$

For each case, determine whether the Markov chain is regular, whether a limiting distribution exists, and whether a stationary distribution exists. Find the $\pi = (\pi_0, \pi_1, \pi_2)^\top$ as a function of p and q .

Solution. Regularity is sufficient, but not necessary, for the existence of a limiting distribution. Thus, if the chain is not regular, we investigate limiting and stationary distributions directly.

- For $0 < p < 1$ and $q = 1 - p > 0$, we see

$$\mathbf{P}^2 = \begin{bmatrix} p^2 & 2pq & q^2 \\ q^2 & p^2 & 2pq \\ 2pq & q^2 & p^2 \end{bmatrix} = \begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \end{bmatrix},$$

meaning the chain is regular. By [Theorem 1.3](#), a unique limiting distribution exists, and it is also the unique stationary distribution. We find it by solving $(\mathbf{P}^\top - \mathbf{I})\pi = \mathbf{0}$ together with $\mathbf{1}^\top \pi = 1$. One of the three stationarity equations is redundant, so we drop the third one. This gives

$$\begin{bmatrix} p-1 & 0 & q \\ q & p-1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} \pi_0 \\ \pi_1 \\ \pi_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Since $1 - p = q$, this simplifies to

$$\begin{bmatrix} -q & 0 & q \\ q & -q & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} \pi_0 \\ \pi_1 \\ \pi_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \implies \begin{bmatrix} -1 & 0 & 1 \\ 1 & -1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} \pi_0 \\ \pi_1 \\ \pi_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix},$$

Within the present case, the limiting probabilities are therefore independent of p and q . Solving the system using Gauss elimination, we find

$$\pi = (1/3, 1/3, 1/3)^\top.$$

- For $p = 1$ and $q = 0$, we have $\mathbf{P} = \mathbf{I}$ and hence $\mathbf{P}^k = \mathbf{I}$ for every $k \geq 1$. The matrix contains zero entries for every k , so the chain is not regular. Every probability distribution is stationary because $\pi = \mathbf{I}\pi$. However, there is no common limiting distribution as the rows of $\mathbf{P}^t = \mathbf{I}$ depend on the initial state.
- For $p = 0$ and $q = 1$, we have

$$\mathbf{P} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}, \quad \mathbf{P}^2 = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \quad \mathbf{P}^3 = \mathbf{I}, \quad \mathbf{P}^4 = \mathbf{P},$$

and these three matrices repeat. Each power has zero entries, so the chain is not regular. The chain moves deterministically through the three states, and the entries of $\mathbf{P}^n$ oscillate. Hence, no limiting distribution exists. The stationarity equation $\mathbf{P}^\top \pi = \pi$ gives

$$\pi_0 = \pi_2, \quad \pi_1 = \pi_0, \quad \pi_2 = \pi_1.$$

Together with $\pi_0 + \pi_1 + \pi_2 = 1$, this shows that the unique stationary distribution is

$$\boldsymbol{\pi} = (1/3, 1/3, 1/3)^\top.$$

Problem 1.3. A Markov chain $\{X_t : t = 0, 1, \dots\}$ has transition probability matrix

$$\mathbf{P} = \begin{bmatrix} 1/2 & 1/2 & 0 & 0 \\ 1/2 & 0 & 1/2 & 0 \\ 0 & 1/2 & 0 & 1/2 \\ 0 & 0 & 1/2 & 1/2 \end{bmatrix}.$$

What is its limiting distribution? What fraction of time, in the long run, does the chain spend in state 0?

Solution. Using the fact that the chain is doubly stochastic, we can from [Theorem 1.7](#) immediately conclude that the limiting distribution is uniform $\boldsymbol{\pi} = (1/4, 1/4, 1/4, 1/4)^\top$. Furthermore, by [Theorem 1.9](#), the long-run fraction of time spent in state 0 is $\pi_0 = 1/4$.

Problem 1.4. (Exam August 2025) On any given day Maria is either cheerful (C), so-so (S), or glum (G). If she is cheerful today, then she will be C, S, or G tomorrow with respective probabilities 0.5, 0.4, 0.1. If Maria is feeling so-so today, then she will be C, S, or G tomorrow with probabilities 0.3, 0.4, 0.3. If she is glum today, then she will be C, S, or G tomorrow with probabilities 0.2, 0.3, 0.5. Let X_n denote Maria's mood on day n .

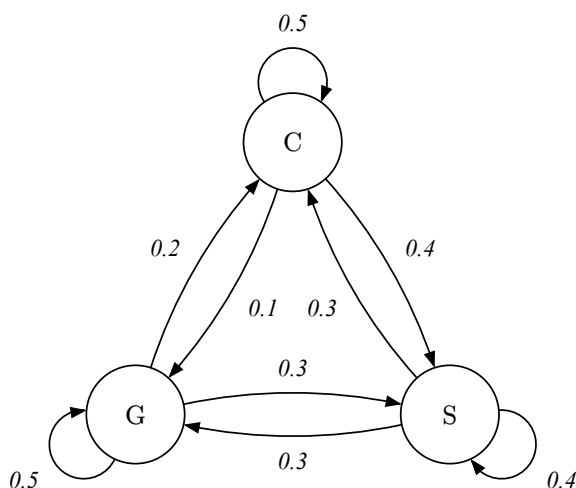
1.
 - Explain that Maria's mood is a three-state Markov chain.
 - Illustrate with a transition probability graph, and determine the transition probability matrix $\mathbf{P}$.
2. Assume that Maria's mood at day 27 has a uniform probability distribution. What is the probability that Maria is glum at day 28.
3.
 - Explain that the Markov chain has a unique stationary distribution.
 - How often, on average, is Maria glum?
 - Is it possible that the distribution of X_1 equals the distribution of X_n for all n ?

Solution.

1.
 - Three states C, S, and G. The Markov property holds since the probability of Maria's mood tomorrow only depends on her mood today.
 - The transition probability matrix is

$$\mathbf{P} = \begin{bmatrix} 0.5 & 0.4 & 0.1 \\ 0.3 & 0.4 & 0.3 \\ 0.2 & 0.3 & 0.5 \end{bmatrix},$$

and the transition probability graph is



2. As mood at day 27 is uniformly distributed, we have

$$P(X_{27} = C) = P(X_{27} = S) = P(X_{27} = G) = 1/3.$$

The probability that Maria is glum at day 28 is therefore

$$\begin{aligned} P(X_{28} = G) &= \sum_{k \in \{C, S, G\}} P(X_{28} = G \mid X_{27} = k) P(X_{27} = k) \\ &= 0.1 * 1/3 + 0.3 * 1/3 + 0.5 * 1/3 \\ &= 0.3. \end{aligned}$$

3. The Markov chain is regular since $P_{i,j} > 0$ and the state space is finite. By [Theorem 1.3](#), it has a unique stationary distribution $\pi = \mathbf{P}^\top \pi$.

- We have the system of equations

$$0.5\pi_C + 0.3\pi_S + 0.2\pi_G = \pi_C,$$

$$0.4\pi_C + 0.4\pi_S + 0.3\pi_G = \pi_S,$$

$$0.1\pi_C + 0.3\pi_S + 0.5\pi_G = \pi_G,$$

$$\pi_C + \pi_S + \pi_G = 1,$$

where we have four equations and three unknowns. Dropping the last equation, we can solve the system

$$\begin{bmatrix} -0.5 & 0.3 & 0.2 \\ 0.4 & -0.6 & 0.3 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} \pi_C \\ \pi_S \\ \pi_G \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \implies \begin{bmatrix} \pi_C \\ \pi_S \\ \pi_G \end{bmatrix} = \begin{bmatrix} 21/62 \\ 23/62 \\ 18/62 \end{bmatrix}.$$

Hence, on average, Maria is glum $\pi_G = 18/62 \approx 29\%$ of the time.

- Yes, this is possible if the initial distribution is the stationary distribution π .

Example 1.10. (PageRank) A search engine should not only check whether a page contains a search term, but also whether the page is *important*. PageRank assigns a score to each page using the link structure of the web. The key idea is to model a random web surfer. At each step, the surfer either follows a hyperlink or jumps to another page. Pages that are visited often in the long run get a high PageRank score. This idea was introduced in the early Google search engine Brin and Page (1998).

Suppose there are n pages. Each page is a state in a Markov chain. If page i has d_i outgoing links, then the hyperlink transition matrix is

$$P_{i,j} = \begin{cases} 1/d_i & \text{if page } i \text{ links to page } j \\ 0 & \text{otherwise.} \end{cases}$$

From page i , the surfer chooses uniformly among its outgoing links. If a page has no outgoing links, it is called a *dangling page*. Its row would contain only zeros, so we replace it by a uniform distribution

$$S_{i,j} = \begin{cases} P_{i,j} & \text{if } d_i > 0 \\ 1/n & \text{if } d_i = 0. \end{cases}$$

This makes $\mathbf{S}$ a transition probability matrix, but does not necessarily make the chain regular. PageRank therefore adds *teleportation*. Let $\alpha \in (0, 1)$ be the probability of following a link, and let $\mathbf{v}$ be a probability distribution with $v_j > 0$ for all j . The PageRank transition matrix is

$$G_{i,j} = \alpha S_{i,j} + (1 - \alpha)v_j.$$

A common choice is $\alpha = 0.85$ and $v_j = 1/n$. Since $G_{i,j} \geq (1 - \alpha)v_j > 0$, $\mathbf{G}$ is regular. It therefore has a unique stationary and limiting distribution $\boldsymbol{\pi}$, which is the PageRank vector. It satisfies $\boldsymbol{\pi} = \mathbf{G}^\top \boldsymbol{\pi}$ and $\mathbf{1}^\top \boldsymbol{\pi} = 1$, and we may compute it from any initial distribution $\boldsymbol{\pi}^{(0)} = (1/n, \dots, 1/n)^\top$ by iterating

$$\boldsymbol{\pi}^{(t+1)} = \mathbf{G}^\top \boldsymbol{\pi}^{(t)}.$$

It describes the long-run fraction of time spent on each page, meaning we want to rank pages based on this ordering.

Boom, now you have a gazillion dollar business idea! See the `lectures/week36/code/pagerank.py` for an implementation of this algorithm.

References

Brin, S., and Page, L. The anatomy of a large-scale hypertextual Web search engine. *Computer Networks and ISDN Systems*, 30(1), 107–117, 1998. [https://doi.org/10.1016/S0169-7552\(98\)00110-X](https://doi.org/10.1016/S0169-7552(98)00110-X)